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SARAVANAKUMAR I

SUMMARY

Electronics communication Engineer with knowledge of Microsoft Azure services, including virtual machines, storage, networking, identity management, and cloud security. Skilled in deploying, monitoring, and managing cloud resources while ensuring reliability, scalability, and performance. Strong problem-solving abilities with a commitment to continuous learning and delivering efficient cloud solutions using Artificial Intelligence.

EDUCATION

ER.PERUMAL MANIMEKALAI COLLEGE OF ENGINEERING, Koneripalli **2022-2026**
Bachelor of Engineering, Electronics and communication 7.0 CGPA

PRIMUS PU COLLEGE, Bengaluru **2020-2022**
2nd PU College

TECHNICAL SKILL

- Hands-on exposure to sensor integration and hardware debugging
- Strong grasp of structured programming and embedded C implementation.
- Project knowledge of Python
- Azure Administration (Basic Knowledge):VM provisioning, Virtual Network (VNet) configuration, Azure AD Identity & Access Management (RBAC), Network Security Groups (NSG), Azure Monitor, and Azure Storage services.
- Knowledge on Azure resource provisioning using terraform & terragrunt
- Technical Skills on Git repository
- Good Technical Skills on Linux Administration with Scripting
- Functional Knowledge of Artificial Intelligence and Generative AI
- Good Skill on Powerpoint slide preperation for Projects
- Knowledge of MS Excel upto Intermediate Level

LANGUAGES

- Tamil(Fluent)
- English(Fluent)
- Kannada(Fluent)
- Hindi(Basic Conversational)

PROJECTS

Smart Power Grid Monitoring and Optimization using Azure & Reinforcement Learning

- Performed signal scaling for safe ESP32 ADC interfacing.
- Acquired real-time voltage data using ESP32.
- Transmitted telemetry to Azure IoT Hub via Wi-Fi.
- Implemented reinforcement learning (Q-Learning) for load optimization logic.
- Analyzed voltage variation under different load conditions.
- Simulated intelligent load control decisions.
- Ensured electrical isolation and safe high-voltage handling.

AI-Based Fire Detection System

- Engineered a real-time computer vision model using YOLO to autonomously detect fire and smoke patterns with high precision.
- Architected an edge-computing solution by integrating the AI model with an ESP32-CAM to facilitate localized processing and IoT-based remote alerting.
- Developed a hardware prototype featuring synchronized LED indicators and buzzer alarms for immediate, localized hazard notification.
- Optimized the system design to provide a low-cost, scalable safety solution suitable for large-scale disaster management applications.

Achievements

- **Inter-College Champion**, Technical Paper Presentation – PSV Engineering College, Bargur
- **Inter-College Champion**, Technical Paper Presentation – Jayalakshmi Institute of Technology, Thoppur
- **Inter-College Runner-Up**, Technical Paper Presentation – Er.Perumal Manimekalai College of Engineering, Koneripalli

CERTIFICATION

- Embedded Systems & IOT -Skill Training Program
- Technical Workshop on Networking and Security
- Foundation Skills IOT – Skill Training Program